YICHONG CHEN

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EDUCATION

Imperial College London *Apr 2022 - present*
Degree: Ph.D (Computing)
Research: AI/ML-driven system management for dynamic distributed edge computing environments.

Imperial College London *Sep 2019 - Nov 2020*
Degree: Master of Science (Computing Artificial Intelligence and Machine Learning)
Awarded: Distinction (80.2%)
Courses: Reinforcement Learning, Deep Learning, Natural Language Processing, Computer Vision.

University of Birmingham *Sep 2017 - Jun 2019*
Degree: Bachelor of Science (Mathematics)
Awarded: First Class Honor (81%)
Courses: Applied Statistics, Non-linear Programming & Heuristic Search, Real and Complex Analysis

South China University of Technology *Sep 2014 - Jun 2019*
Degree: Bachelor of Engineering (Information Management and Information System)
Courses: Data Mining and Statistics, Object-oriented Programme Design, Operational Research

PROFESSIONAL EXPERIENCE

Ph.D. Researcher, Imperial College London *Apr 2022 – Present*

- Research focuses on designing AI&ML methods to optimize quality of service (QoS) in large-scale distributed systems, enabling deployment of powerful AI models on edge devices.
- Identified and formalized critical challenges in deploying AI models to edge devices, motivating the need for more effective deployment and optimization techniques.
- Independently developed a distributed AI edge system with adaptive neural networks to reduce inference time and energy use on resource-constrained devices while maintaining model performance.
- Designed a novel optimization algorithm integrating a transformer-based neural performance predictor and a GNN-based system-performance estimator to optimize accuracy and latency for multi-device inference jointly.
- Developed a context-aware Reinforcement Learning model to auto-reconfigure distributed system settings in real time. Bridging experimental assumptions with realistic dynamic edge environments and addressing practical system management challenges.
- Built an AI-driven scheduler to continuously optimize task–host placement in a large-scale fog-computing environment, reducing response time by 8% and service level agreement violations by 50%.
- Defined research milestones and a publication roadmap in collaboration with supervisors and international collaborators, leading to first-authored publications at venues such as INFOCOM, CNSM, and MASCOTS.

AI Engineer, Huawei Technologies Co., Ltd *Nov 2020 - Jun 2022*

- Designed and implemented scalable data pipelines using Apache Spark and Hadoop to collect, clean, and process massive datasets from industrial production lines.
- Built a real-time manufacturing data monitoring system and ensemble anomaly-detection pipelines to identify early quality issues on production lines, reducing maintenance costs by 30%.
- Led and collaborated with cross-functional engineers to define standardized industrial production issue-analysis procedures and build a labeled problem-analysis dataset.
- Designed and trained a transformer-based language model to recommend solutions to production-line analysts, reducing issue resolution time by 20%.
- Identified efficiency bottlenecks in business workflows and designed an OCR-based text-extraction framework to pull key values from oscilloscope images, reducing engineers' manual data entry by 90%.

- Developed clean, well-structured, and maintainable codebases, enabling sustainable evolution and easy extension of related services.

Full Stack Developer, Imperial Consultants *Sep 2022 – Oct 2024*

- Led end-to-end design, development, and cloud deployment of SiMON, a web-based solar-field anomaly monitoring platform.
- Built full-stack solution: React (frontend), Python (backend), and a MATLAB-based algorithm service containerized with Docker. Configured Nginx as a reverse proxy/load balancer.
- Worked directly with client stakeholders at SEENG LTD to collect and prioritize requirements, define the product roadmap, and iteratively refine features based on feedback.
- Delivered a production-ready system that enabled SEENG LTD to demonstrate business value and win new funding and external contracts.

TECHNICAL SKILLS

Programming Languages	Python, MATLAB, JavaScript
Frameworks & Tools	PyTorch, FastAPI, Flask, React, Vue3
Languages	English (Business Conversation), Mandarin (Native), Cantonese (Native)
Website	https://chanyikchong.github.io/

PUBLICATIONS

Y. Chen, Z Niu, M Roveri, G Casale. CEED: Collaborative Early Exit Neural Network Inference at the Edge, in *Proc. of INFOCOM*, May 2025.

S. Wang, K Li, D You, **Y Chen**, A Lin, S Liu, X Li, JA McCann. Wimans: A benchmark dataset for wifi-based multi-user activity sensing , in *Proc. of ECCV*, Oct 2024.

Y. Chen, M Roveri, S Tuli, G Casale. Coupling QoS Co-Simulation with Online Adaptive Arrival Forecasting, in *Proc. of IFIP/IEEE CNSM*, Nov 2023.

Y. Chen, G. Casale. Deep Learning Models for Automated Identification of Scheduling Policies, in *Proc. of IEEE MASCOTS*, Nov 2021.